Blender PBR Material Shading Self Learning Guide

Learning outcomes

- Explain physically based rendering and the purpose of each texture map.
- Download and identify the maps in a PBR texture set.
- Connect colour and data maps using correct colour spaces.
- Build normal and true displacement workflows.
- Control one coordinate system across all maps.
- Create the complete material automatically with Node Wrangler.

Before starting

1.Be familiar with Material Properties, Shader Editor, Principled BSDF and Node Wrangler.
2. Open Blender and save the file as StudentName_PBR_Material_Part2.blend. 3. Use a UV sphere or a cube with bevelled edges as the practice object.
4. Switch to the Shading workspace and Material Preview or Rendered view.
5.Enable Node Wrangler from Edit > Preferences > Add-ons or Extensions.

1 Understand PBR Materials

PBR means Physically Based Rendering. A PBR material uses several coordinated images to describe different physical properties of the same surface. When all maps use the same UV coordinates, Blender can calculate a more believable interaction between the material and light.

Map	Controls	Typical appearance	Colour space
Base Color or Albedo	Visible surface colour	Full colour photograph without strong lighting	sRGB
Specular or Reflectivity	Strength or tint of reflections	Usually grayscale; sometimes coloured	Non-Color
Roughness	How blurred reflections are	White rough; black smooth	Non-Color
Glossiness	Smoothness; inverse of roughness	White smooth; black rough	Non-Color then invert
Normal	Small bumps and surface directions	Blue-purple image	Non-Color through Normal Map
Height or Displacement	Physical or apparent surface height	Grayscale height information	Non-Color
Ambient Occlusion	Darkening in creases	Grayscale contact-shadow detail	Non-Color; optional mix with base colour

Core rule. Never connect every map directly to Base Color. Each image describes a different property and must go to the matching input.

2 Download and Inspect a PBR Texture Set

6. Open a legitimate PBR texture website and select one material such as stone, brick, wood or concrete.
7. Download a free texture set at a practical resolution such as 1K or 2K for classroom work. 8.Extract the downloaded ZIP file into a clearly named folder.
9. Do not rename individual map files until you identify their purpose.
10. Look for names containing albedo, basecolor, diffuse, color, specular, roughness, gloss, normal, height, displacement or AO.
11. Confirm that all images belong to the same material and have matching dimensions where possible.
12. Keep the texture folder beside the Blender file or use File > External Data > Pack Resources after setup.

Expected result. A single folder contains the coordinated images for one material.

3 Prepare the Practice Object

13. Select the default cube or add a UV Sphere using Shift A > Mesh.
14. If using a cube, bevel its edges slightly so reflections are visible.
15. Press Ctrl A > Scale to apply object scale. Unapplied scale can make procedural coordinates and displacement harder to control.
16. Create a new material named PBR Practice.
17. Open the Shader Editor and confirm that Principled BSDF is connected to Material Output > Surface.
18. For accurate image placement, enter Edit Mode, select all faces with A and unwrap with U > Unwrap or Smart UV Project.

4 Connect the Base Colour Map

19. Press Shift A > Texture > Image Texture.
20. Click Open and select the Base Color, Albedo, Diffuse or Color image.
21. Keep Color Space set to sRGB because this image is intended to be viewed as colour.
22. Connect Image Texture Color to Principled BSDF Base Color.
23. Switch to Material Preview and check that the visible colour pattern appears on the object. 24.

If the material is stretched, complete the mapping and UV steps later in this guide. **Checkpoint 1.** The object shows the material's colour, but it may still look flat because the physical property maps are not connected yet.

5 Connect Specular or Reflective Information

25. Add another Image Texture node and open the Specular, Reflection or Reflectivity map if the

texture set provides one.

26. Set Color Space to Non-Color because this image supplies numerical data rather than display colour.
27. In older Principled BSDF versions, connect Color to Specular. In current Blender versions, connect it to Specular IOR Level only when the map is intended for that workflow. 28. Observe how brighter map regions receive stronger specular response.
29. If the downloaded material follows a metallic workflow instead, connect the Metallic map to Metallic and keep it Non-Color.

30. Do not connect a metallic map to Specular merely because both influence reflections; they describe different material models.

Important. Many modern texture sets use Metallic plus Roughness rather than a dedicated Specular map. Use only the maps supplied for the same workflow.

6 Add Roughness or Gloss

6.1 When a roughness map is supplied

31. Add an Image Texture and open the Roughness map.
32. Set Color Space to Non-Color.
33. Connect Color to Principled BSDF Roughness.
34. Inspect the result under a light or HDRI: white regions should look rougher and black regions smoother.

6.2 When only a gloss map is supplied

35. Add an Image Texture and open the Gloss or Glossiness map.
36. Set it to Non-Color.
37. Add an Invert Color node between the gloss image and Principled BSDF Roughness.
38. Connect Gloss Color to Invert Color, then Invert Color to Roughness.
39. Remember that gloss is the inverse of roughness: white gloss means smooth, while white roughness means rough.

Checkpoint 2. Reflections vary across the surface instead of using one uniform roughness value.

7 Add the Normal Map Correctly

40. Add an Image Texture and open the Normal map. It normally appears blue or purple.
41. Set Color Space to Non-Color.
42. Add a Normal Map node using Shift A > Vector > Normal Map.
43. Connect Image Texture Color to Normal Map Color.
44. Connect Normal Map Normal to Principled BSDF Normal.
45. Adjust Strength carefully. Begin around 1.0 and reduce it if the bumps look exaggerated. 46. If bumps appear pushed inward when they should project outward, check whether the map is OpenGL or DirectX. Blender normally expects OpenGL orientation; invert the green channel only when required.

Expected result. The surface appears detailed under changing light, but the object's silhouette and geometry remain unchanged.

8 Add Height and Displacement

8.1 Bump only method

47. Add an Image Texture and open the Height or Displacement map.
48. Set Color Space to Non-Color.
49. Add a Bump node.
50. Connect the height image Color to Bump Height.
51. Connect Bump Normal to Principled BSDF Normal.
52. If a Normal Map is already present, connect Normal Map Normal to Bump Normal so both types of detail are combined.

53. Adjust Bump Strength and Distance conservatively.

8.2 True displacement method in Cycles

54. Change the render engine to Cycles.
55. Add sufficient geometry using subdivision or a Subdivision Surface modifier. Displacement cannot create detail without enough mesh subdivisions.
56. Add a Displacement node.
57. Connect the height image Color to Displacement Height.
58. Connect Displacement output to Material Output > Displacement.
59. Set Midlevel near 0.5 when middle grey represents the unchanged surface.
60. Adjust Scale gradually; large values can explode or distort the mesh.
61. In Material Settings, choose Displacement and Bump or the current equivalent when true displacement is required.

Method	Changes silhouette	Speed	Best use
Normal map	No	Fast	Fine scratches and small surface detail
Bump map	No	Fast	Grayscale height used as lighting detail
True displacement	Yes	Slower	Large bricks, stones and visible surface relief

Decision rule. Use normal or bump for small detail. Reserve true displacement for geometry that must affect the outline or cast real shadows.

9 Add Consistent Texture Mapping

62. Select any Image Texture node.
63. Add Texture Coordinate and Mapping nodes, or press Ctrl T with Node Wrangler enabled.
64. Connect Texture Coordinate UV to Mapping Vector.
65. Connect Mapping Vector to the Vector input of every Image Texture node. 66. Use the same Mapping output for all maps so base colour, roughness, normal and displacement remain aligned.
67. Adjust Mapping Location and Rotation only when the entire PBR set must move together. 68. If

image placement must follow a particular mesh layout, edit the object's UV map rather than independently moving one texture.

Core rule. Every PBR image must use matching coordinates. If only one map is shifted or scaled differently, the surface properties will no longer correspond.

10 Adjust PBR Texture Scale

69. Use the Mapping node Scale values to change the repetition of the whole texture set. 70. Increase X, Y and Z scale values equally to produce more repetitions and smaller visible features.
71. Decrease them equally to produce fewer repetitions and larger visible features.
72. For UV mapping, you can instead scale UV islands in the UV Editor.
73. Compare the visible brick, wood grain or stone size with the real-world size of the object.
74. Do not judge scale from a close-up alone; zoom out and inspect the full object. 75. Apply the same scale to every map by sharing one Mapping node.

Expected result. Texture scale should communicate believable real-world size. A brick should not become the size of an entire building wall unless intentionally stylized.

11 Faster Setup with Node Wrangler

76. Confirm that Node Wrangler is enabled.
77. Select the Principled BSDF node.
78. Press Ctrl Shift T. This opens the Principled Texture Setup file browser.
79. Navigate to the extracted PBR texture folder.
80. Select the related Base Color, Roughness or Gloss, Normal, Displacement or Height, Metallic and AO images together.
81. Click Principled Texture Setup.
82. Inspect the generated nodes. Node Wrangler normally connects recognized files, applies suitable Non-Color settings to data maps and adds Mapping and Texture Coordinate nodes. 83. Verify every connection manually. File naming can cause a map to be misidentified. 84. If a Gloss map was connected directly to Roughness, insert an Invert node. 85. If displacement is not desired, disconnect it or use the height map through a Bump node instead.

12 Improve and Organize the Node Tree

86. Arrange data flow from left to right: Texture Coordinate, Mapping, Image Textures, conversion nodes, Principled BSDF and Material Output.
87. Place related nodes inside labelled Frames using Ctrl J.
88. Label frames Coordinates, Surface Maps and Height Detail.
89. Rename image nodes or add labels such as BASE COLOR, ROUGHNESS, NORMAL and HEIGHT.
90. Use Ctrl Shift click with Node Wrangler to preview one map at a time.
91. Use Ctrl plus right-drag to cut incorrect links.
92. Save the Blender file with Ctrl S.
93. Use File > External Data > Pack Resources if the file must work on another computer without the original texture folder.

13 Complete Connection Reference

Texture image	Required intermediate node	Final connection
Base Color or Albedo	None	Principled BSDF Base Color
Roughness	None	Principled BSDF Roughness
Glossiness	Invert Color	Principled BSDF Roughness
Metallic	None	Principled BSDF Metallic
Specular	Workflow dependent	Principled BSDF Specular IOR Level or older Specular input
Normal	Normal Map	Principled BSDF Normal
Height for bump	Bump	Principled BSDF Normal
Height for true displacement	Displacement	Material Output Displacement

Ambient Occlusion	Mix or Multiply with Base Color when appropriate	Principled BSDF Base Color

14 Student Practice Activity

94. Choose one PBR material: brick, stone, wood or concrete.
95. Apply it to a cube, sphere or simple wall model.
96. Connect Base Color, Roughness or Gloss, Normal and Height maps manually.
97. Create a second copy of the material using Ctrl Shift T.
98. Compare the manual and automatic node trees.
99. Adjust texture scale until it matches the object's believable size.
100. Create one render using only normal or bump detail.
101. Create a second Cycles render using true displacement when the computer permits.
102. Write three sentences explaining which version is more suitable and why.

15 Submission Requirements

- One .blend file with clearly named objects and materials.
- The downloaded PBR texture set or packed Blender resources.
- Screenshot of the complete manual node setup.

• Screenshot of the Node Wrangler generated setup.
• Rendered image showing colour, reflection variation and surface detail.
• Short comparison of normal, bump and displacement.

Quick Shortcuts

Shortcut	Purpose
Shift A	Add a node
Ctrl T	Add Texture Coordinate and Mapping to a selected texture
Ctrl Shift T	Build a Principled PBR texture setup from selected maps
Ctrl Shift click	Preview a selected node output
Ctrl plus right-drag	Cut node links
Ctrl J	Frame selected nodes
Ctrl A > Scale	Apply object scale
U	Open UV unwrap menu in Edit Mode
Ctrl S	Save